View Report

R1

(Number of First Attempts: 87)

HW2

Question 1 Difficulty: 1

What is the identifying assumption in Regression Discontinuity Design (RDD)?

Average Grade: 0.94 / 1 (93.81 %)

Treatment is able to be assigned randomly without regulatory / ethical considerations	1 (1.03 %)
➡ Ex ante equivalence between subjects within bandwidth of cutoff, regardless of whether they are below or above cutoff	91 (93.81 %)
Non zero correlation between treatment incidence and error term from the estimation involving y (ultimate dependent variable) regardless of whether in 1 or 2 stages	1 (1.03 %)

All of the options	4	(4.12 %)

Question 2 Difficulty: 1

What is the independent variable of interest in the second stage of 2SLS

Observed value of x	0	(0 %)
➡ Predicted value of x, which is obtained from the first stage of 2SLS	95	(97.94 %)
Predicted value of x, which is obtained from the second stage of 2SLSS	2	(2.06 %)
Lagged values of y (LHS variable in the second stage)	0	(0 %)

Average Grade: 0.98 / 1 (97.94 %)

Question 3 Difficulty: 1

How does a valid instrumental variable affect y (ultimate dependent variable in 2SLS)?

Average Grade: 0.95 / 1 (94.85 %)

It must be strictly uncorrelated to both y and all other RHS variables in the first stage of 2SLS only	0	(0 %)
It must be strictly uncorrelated to both y and all other RHS variables in both stages of 2SLS	1	(1.03 %)

It will affect y via its effect on x (dependent variable in the first stage of 2SLS), and it does not matter if it affects y through other channels	4 (4.12 %)
➡ It will only affect y via its effect on x (dependent variable in the first stage of 2SLS), and strictly through no other channels	92 (94.85 %)

Question 4 Difficulty: 1

What are some examples of instrumental variable categories which are commonly used in the corporate finance literature?

Average Grade: 0.99 / 1 (98.97 %)

Changes in regulated accounting policy by national or supranational accounting authorities	1 (1.03 %)
Natural phenomenon such as weather patterns, storms, etc	0 (0 %)
Changes in governmental policy (especially if	0 (0 %)

there is cross sectional dispersion)

➡ All of the options are valid — 96 (98.97 %)

Question 5 Difficulty: 1

We have panel data on stock returns, and also measures of changes in corporate compensation policy for a set of US listed firms from 2010 to 2024. Can we just use OLS to regress "stock returns (T)" against "measure of corporate compensation policy(T-1)" to determine the causal impact of compensation policy on stock returns?

Average Grade: 0.97 / 1 (96.91 %)

➡ No. Because corporate compensation policy (and generally speaking all corporate policies) are endogeneous. It will be unclear from the estimation which direction causality is running in, and also impossible to eliminate the existence of a 3rd mediating / causal variable — 94 (96.91 %)

No. Because corporate compensation policy (and generally speaking all corporate policies) are exogeneous, so we should use the — 3 (3.09 %)

compensation data as an instrumental variable in 2SLS		
Yes. Because corporate compensation policy (and generally speaking all corporate policies) are exogeneous, so we can use a simple panel regression, potentially using fGLS to cluster our standard errors	0	(0 %)
Yes. Because time only flows in 1 direction, if there is a statistically significant relationship, it must be because compensation policy has a causal impact on returns	0	(0 %)

Question 6 Difficulty: 1

Are the confidence intervals from instrumental variables estimation via 2SLS always more precise than OLS?

Average Grade: 0.97 / 1 (96.91 %)

Yes, but only if there exists more than 1 instrumental variable	0	(0 %)
Yes, they are generally	0	(0 %)

narrower than OLS. If there exist a valid instrument, we should always prefer IV via 2SLS over OLS even if it's a weak instrument

No, they are generally wider than OLS, but IV estimation via 2SLS is consistent in situations where OLS will be inconsistent due to violation of assumption A4 — 94 (96.91 %)

No, they are generally wider than OLS, and also IV estimation via 2SLS is inconsistent in situations where OLS will is consistent and vice versa — 3 (3.09 %)

Question 7 Difficulty: 1

When can we perform overidentifying test of uncorrelatedness between instrument and error term from 2SLS?

Average Grade: 0.93 / 1 (92.78 %)

When we have exactly 1 instrument only — 1 (1.03 %)

When we have strictly more — 90 (92.78 %)

than 1 instrument		
When we have strictly more than 2 instruments	6	(6.19 %)
We can always do it regardless of number of instruments	0	(0 %)

Question 8 Difficulty: 1

In RDD, if there is differential density in observations just below the cutoff (but within bandwidth) compared to just above the cutoff (but within bandwidth), what is one possible objection to using RDD?

Average Grade: 1 / 1 (100 %)

Subjects are able to manipulate their treatment status, meaning another (potentially unobserved) variable is responsible for differences in density. Such a variable may also affect outcomes, and it will not be possible to differentiate this from the effect of treatment	97	(100 %)
This directly implies same cutoff on assignment value is used to determine multiple treatment modalities, and we will not be able to differentiate between the effect of each of the treatments on the outcome variable	0	(0 %)
This implies that probability of treatment does not	0	(0 %)

change monotonically with score on the assignment variable

None of the options 0 (0 %)

Question 9 Difficulty: 1

What is the main difference between sharp RDD and fuzzy RDD?

Average Grade: 0.99 / 1 (98.97 %)

- In fuzzy RDD, probability of treatment does not have to change from exactly 0.0 to 1.0 following a crossing of the cutoff — 96 (98.97 %)
- In sharp RDD, probability of treatment does not have to change from exactly 0.0 to 1.0 following a crossing of the cutoff — 1 (1.03 %)
- In fuzzy RDD, probability of treatment does not have to be monotonically changing as value of the assignment variable increases — 0 (0 %)
- In sharp RDD, probability of treatment does not have to be monotonically changing as value of the assignment — 0 (0 %)

variable increases

Question 10 Difficulty: 1

In our class case study regarding Russell index rebalancing, why is there short term abnormal positive returns for companies which drop from bottom of Russell 1000 to top section of the Russell 2000 index?

There is significantly less AUM tracking Russell 2000 compared to Russell 1000	3 (3.09 %)
There is significantly more AUM tracking Russell 2000 compared to Russell 1000	5 (5.15 %)
➡ Both indices are value weighted; there is order of magnitude more index weight for companies at the top of Russell 2000 compared to bottom of Russell 1000, while only incrementally more AUM tracks Russell 1000 compared to Russell 2000	88 (90.72 %)
None of the options	1 (1.03 %)

Average Grade: 0.91 / 1 (90.72 %)

Question 11 Difficulty: 1

How do we estimate Fuzzy RDD?

➡ Via 2SLS	97	(100 %)
Via OLS	0	(0 %)
Via Lasso estimation	0	(0 %)
Via Ridge regression	0	(0 %)

Average Grade: 1 / 1 (100 %)

Question 12 Difficulty: 1

If you had to pick one of the below options, which is the critical bottleneck in IV estimation via 2SLS?

➡ Conceptualizing one or more instrumental variables	82	(84.54 %)
Quantifying correlation between the instrumental variables and the x variable	10	(10.31 %)
Maximizing correlation between the instrumental variables and the y variable	0	(0 %)
All of the options	5	(5.15 %)

Average Grade: 0.85 / 1 (84.54 %)